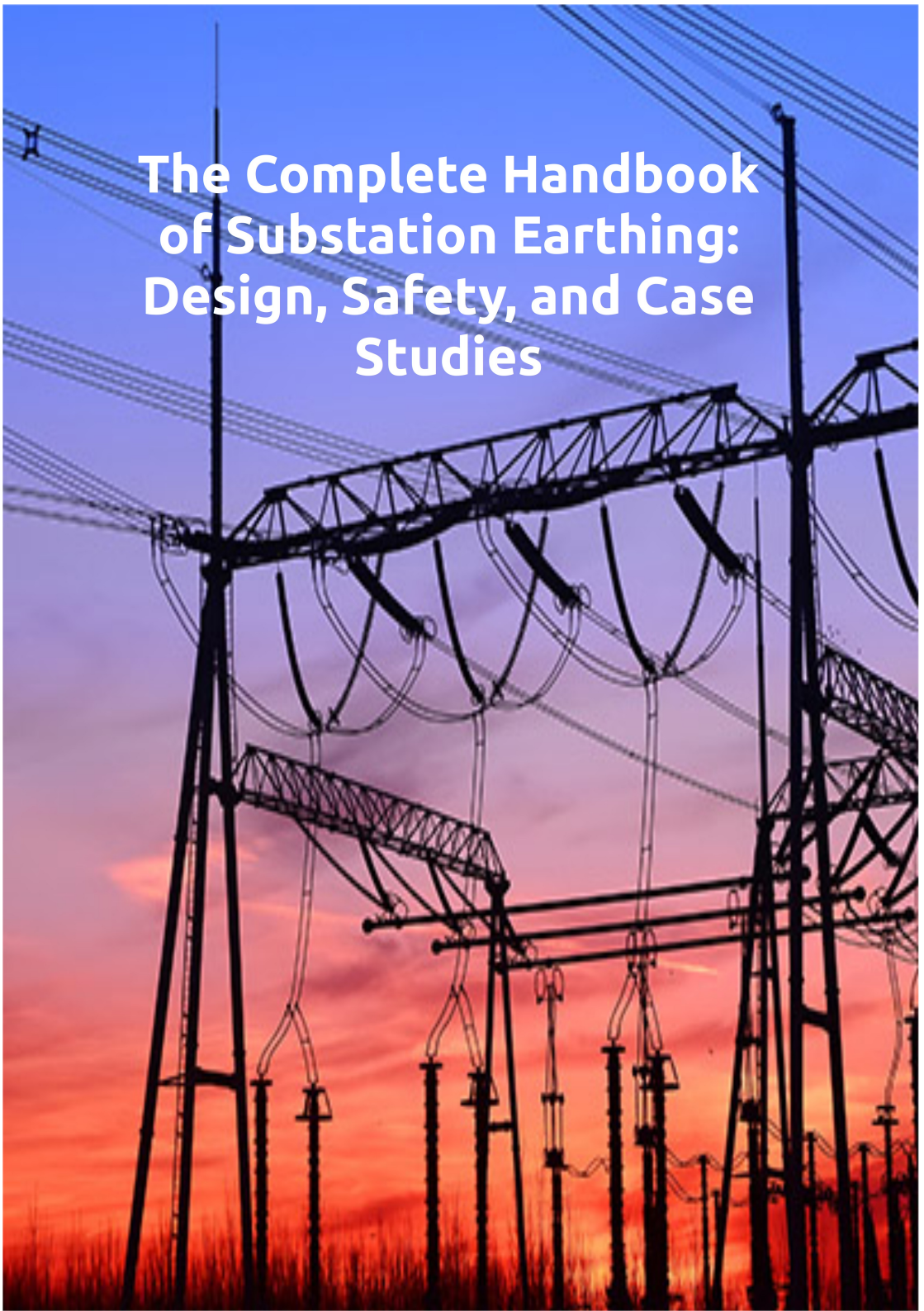


The Complete Handbook of Substation Earthing: Design, Safety, and Case Studies



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1. Definitions

In this document, the following words and expressions will have the following meanings:

Item	Meaning
I	RMS value of earth fault current (kA)
Cu	Earthed electrode in copper
TCAP	Thermal capacity per unit volume ($\text{J}/\text{cm}^3/^\circ\text{C}$)
Tc (ts)	Fault current withstand capacity per conductor ($^\circ\text{C}$)
ar	Thermal resistivity coefficient at the reference temperature ($^\circ\text{C}$)
rr	Resistivity of the earth conductor at the reference temperature ($\mu\text{Ohm-cm}$)
Ko	Reciprocal of the thermal coefficient of resistivity ($^\circ\text{C}$)
Tm	Maximum admissible temperature ($^\circ\text{C}$)
Qf	Final temperature ($^\circ\text{C}$)
Ta	Ambient temperature ($^\circ\text{C}$)
p	Soil resistivity (Ohm-m)
Ca	Corrosion allowance (mm^2)
Kf'	Division factor
V	System voltage
ps	Resistivity of the upper layer (Ohm-m)
Ig	Maximum fault current considered (A)
A	Section area of conductor (mm^2)
d	Equivalent diameter (mm)
L	Length of switchyard (m)

Item	Meaning
B	Width of switchyard (m)
Ag	Equivalent area of switchyard (m ²)
Ds	Conductor spacing (m)
Nx	Number of grids in the X direction
Ny	Number of grids in the Y direction
Lc	Total length of conductors (m)
H	Depth of burial (m)
hs	Depth of the upper layer (m)
n0	Weighting factor of the correction factor

SEV1 to 7 Soil resistivity surveys

2. Purpose

The purpose of this document is to describe the functional requirements for Substation Earthing and their integration into a substation.

3. Scope

This document states the functional requirements regarding Substation Earthing and integration into a substation, as well as items that need to be considered in the area surrounding the substation to ensure the safety of personnel and the general public. The scope includes all items in the field such as soil resistivity, ground resistance, rods, electrodes, potential risers, touch voltage, step voltage, etc.

4. Referenced Documents

The table below lists applicable legislation, standards, and referenced documents.

Standards	Description
IEEE Std. 80/2000	IEEE Guide for Safety in Grounding of Substations
ASTM G57-06	American Society of Testing and Materials Guide for Measuring Earth Resistivity, Ground Impedance, and Earth Surface Potentials of a Grounding System
IEEE Std. 81:2012	Standard for Qualifying Permanent Connections Used in Substation Grounding
IEEE 837:2014	
AS/ISO 1000:1998	The International System of Units (SI) and its Applications
IEC/TS 60479-1:2016	Effect of Current on Human Beings and Livestock - Part 1 General Aspects
IEC/TR 60479-5:2007	Effect of Current on Human Beings and Livestock - Part 5 Touch Voltage Threshold Values for Physiological Effects
CIGRE Technical Brochure 535:2013	EMC within Power Plants and Substations

5. Functional Requirements

The functional requirements of an earthing and bonding system must ensure the following:

- Safety of people
- Protection of equipment
- Operational security

The functional requirements must satisfy the latest edition of IEEE 80-2000 and the listed standards.

5.1 Design Requirements

Input parameters for the design requirements include:

- Soil model in accordance with IEEE-81
- Fault level and contributions
- Fault clearing time
- Fault current

Designing an earthing system typically involves software tools that can simulate electrical grounding characteristics and ensure safety and compliance with standards. Here's a list of both open-source and commercial software commonly used for this purpose:

Open-Source Software:

- CDEGS (Current Distribution, Electromagnetic Fields, Grounding, and Soil Structure Analysis)
- EMTP-RV (Electromagnetic Transients Program)

- Neplan

Commercial Software:

- ETAP (Electrical Transient Analyzer Program)
- SKM Systems Analysis (Power Tools)
- PSCAD (Electromagnetic Transients including DC)
- Amtech ProDesign
- SAFE (Software for the Analysis of Faulted Electrical Networks)

Hybrid (Both Open and Commercial):

- MATLAB/Simulink
- DIgSILENT PowerFactory

In this study, we limit ourselves to the calculation of the substation earthing without simulation. A case study follows, based on Smath software.

5.2 Design Deliverables

The final earth grid design for a specific project must be detailed in an earth grid layout drawing showing the following details as a minimum:

- Conductor type, size, and depth of burial
- Gravel/bitumen thickness and design resistivity
- Ground electrode detail including location, depth, and length
- Earthing risers to the equipment and operator loops
- Earth grid design impedance
- Fault current level in which the earthing system is deemed safe and functional for its purpose

5.3 Constructability Requirements

Earthing systems involve the installation of horizontal and vertical electrodes buried or driven into the general mass of the earth. Key requirements include:

- Backfill material must surround the conductor by at least 150 mm and must be between pH levels of 6 and 10. The resistivity of fill material must be similar to or less than that of the natural soil.
- Vertical electrodes must be driven in using appropriate tools. If they cannot be directly driven, they must be installed in 75 mm boreholes, which should be backfilled with a 50/50 bentonite and gypsum slurry mix.

5.4 Operability Requirements

The earthing and bonding system must operationally:

- Provide a low resistance path to remote earth to ensure fault currents are directed to ground and dissipated for all earthed items during the fault duration.
- Limit the earth potential rise to reduce the likelihood of hazardous voltage exposure to personnel.
- Maintain all conductive plant and equipment at the same potential.
- Not create potentially hazardous conditions to external third-party metallic structures.
- Provide a low impedance path to earth for lightning strikes.
- Offer two independent earthing connections to equipment structures, each capable of carrying the maximum expected fault current for the longest expected clearing time.

6. Case Studies applications

The document presents a case study that includes the calculation of weighted average soil resistivity, ground network check, and three variants of earth grid design based on different soil resistivity values. The calculations cover conductor cross-section area, design inputs, tolerable touch and step potentials, grounding resistance, earth potential rise, mesh voltage, and step voltage.

The case study provides detailed calculations and the resulting design parameters, such as soil resistivity, ground grid section area, conductor spacing, number of electrodes, electrode length and diameter, and the final resistance and voltages.

6.1 Calculation of Weighted Average Soil Resistivity

The following table summarizes the surveys used (SEV 1 to 7) and the results of measuring soil resistivity at different layers. The ground resistivity is non-uniform, and the arithmetic average per sample cannot be used. Therefore, we opted for the weighted average to calculate the resistivity of each survey.

		(Ohm-m)	(m)
SEV1	1	543	1,5
	2	283	3
	3	65,02	10,2
	4	INFINITY	
SEV2		8199,6	0,5
	12	279,2	2
	3	268,6	22,5
	4	INFINITY	
SEV3	1	982,3	0,5
	2	807,3	10
	3	114,5	
SEV4	1	7570	0,5
	2	3270	4
	3	512,1	7
	4	INFINITY	
SEV5		10713,1	0,65
	12	4320,4	15
	3	27,2	
SEV6	1	354,3	3,3
	2	36,62	10,93
	3	4314,4	
SEV7	1	2057,4	2,65
	2	5326,7	7,3
	3	309,9	12,37
	4	49650,3	

Weighted Average Resistivity:

Weighted Average $\rho_w = \Sigma(\rho_i * h_i) / \Sigma h_i$

Where :

- ρ_w is the weighted average soil resistivity (in $\Omega\text{-m}$)
- ρ_i is the resistivity of the i-th soil layer (in $\Omega\text{-m}$)
- h_i is the thickness of the i-th soil layer (in meters)
- Σ represents the summation over all the soil layers

Geometric Mean of Resistivity:

Geometric Mean $\rho_{gm} = e^{((1/n) * (\ln \rho_1 + \ln \rho_2 + \dots + \ln \rho_n))}$

Where :

- ρ_{gm} is the geometric mean of soil resistivity (in $\Omega\text{-m}$)
- $\rho_1, \rho_2, \rho_3, \dots, \rho_n$ are the individual soil resistivity values (in $\Omega\text{-m}$)
- n is the total number of soil resistivity values
- e is the base of the natural logarithm (approximately 2.718)
- $\ln$ represents the natural logarithm function

This formulation of the geometric mean is particularly useful when working with data that has a wide range of values, as it can help to mitigate the influence of outliers or extreme values.

DEPTH	RESISTIVITY SECTION SOUNDING			Weighted average resistivity (*)	LOG(average resistivity weighted)
		(Ohm-m)	(m)	Ohm-m	Ohm-m
SEV1	1	543	1,5		
	2	283	3		
				158,28	5,06
	3	65,02	10,2		
	4	INFINITY			
SEV2		8199,6	0,5		
	1 2	279,2	2		
				428,07	6,06
	3	268,6	22,5		
	4	INFINITY			
SEV3	1	982,3	0,5		
	2	807,3	10		
				815,63	6,70
	3	114,5			
	4	INFINITY			
SEV4	1	7570	0,5		
	2	3270	4		
				1 778,23	7,48
	3	512,1	7		
	4	INFINITY			
SEV5		10713,1	0,65		
	1 2	4320,4	15		
	3	27,2		4 585,91	8,43
SEV6	1	354,3	3,3		
	2	36,62	10,93		
	3	4314,4		110,29	4,70
SEV7	1	2057,4	2,65		
	2	5326,7	7,3		
				2 158,18	7,68
	3	309,9	12,37		
	4	49650,3			
			LOG(Arithmetic mean)		6,59
			Geometric Mean Resistivity (Ohm-m) (**)		726,94

Either the resistivity is ~727 Ohm-m,

6.2 Ground Grid Check Note

6.2.1 Variant 1: Sizing the Earth Grid Based on Resistivity Value

After the calcul the geometric average of resistivity, the resistivity to use in this Variant is $\rho=727 \text{ Ohm-m}$.

1.Design Inputs for Earth Mat Calculation:

RMS value of earth fault current I: (kA)	$I= 25$
Conductor type: Annealed soft drawn Copper	Cu
Thermal capacity per unit Volume TCAP: (J / cm ³ / °C)	TCAP=3,42
Fault current withstand capacity per tc conductor: (s)	$t_c=1$
Thermal resistivity coefficient at the reference temperature at 20 oC ar: (°C) ⁻¹	$\alpha_r=0.00393$
Resistivity of the earth conductor at the reference temperature at 20 °C, rr: (μOhm-cm)	$\rho_r=1,72$
Reciprocal of the thermal coefficient of resistivity Ko: (°C)	$K_0=234$
Maximum admissible temperature	$T_m=1083$

in oC Tm: (°C)	
Final temperature Qf: (°C)	Qf=450
Ambient temperature in °C Ta: (°C)	Ta=50
Soil resistivity P: (Ohm-m)	P=727
Corrosion allowance: 1	1

2.CALCULATION OF CONDUCTOR CROSS-SECTION AREA (IEEE-80)

Conductor cross section	$A_{mm^2} = I \sqrt{\frac{1}{\left(\frac{TCAP \cdot 10^{-4}}{t_c \alpha_r \rho_r} \right) \ln \left(\frac{K_o + T_m}{K_o + T_a} \right)}}$ section=(A_{mm²}) =89.7336
Cross section of the conductor after taking into account the correction tolerance: 89.734 *1=89.734 mm ² and the standard type of cable.	A=95 mm ²
Conductor size selected for Main Earthmat design (mm²) The fault current will be divided into two ways (Kf'= 0.5) in the earth grid. for more safety we will take Kf =0.6	Sg=section *Kf= 53.8402mm ²

3.DESIGN INPUTS FOR EARTH MAT CALCULATION

System voltage V: 110 kV	$n_e := \sqrt{N_x \cdot N_y} :$
System grounding:	Solid
Soil resistivity P: Ohm-m	727
Resistivity of the upper layer Ps: Ohm-m (typical soil to gravel value)	12500
RMS value of fault current is: 25 kA	is=25
Maximum Fault current considered Ig: (A):	$I_g = i_s \cdot K_f \cdot 10^3 = 15000$
Section area X of conductor A: mm ²	A=95
Equivalent diameter d: mm	=0,011
Length of Switchyard (m)	L=66
Width of Switchyard (m)	B=51
Equivalent area of Switchyard (m ²)	Ag= 3366
Conductor spacing Ds: (m)	Ds=2,5
No: grids in the X, Nx direction:	$N_x = (L/D_s) = 26$
No: grids in the Y, Ny direction:	$N_y = (B/D_s) = 20$
Equivalent number of grids ne :	ne=228035
Total length of conductors Lc: m	$L_c = N_x \cdot B + N_y \cdot L = 2646$
Depth of burial H (m)	H=0,8

Depth of the upper layer hs(m)	Hs=0,15
No. of electrodes, Total number of electrodes m:	m=100
Electrode length l0 (m)	l0=10
Total length of the L R electrodes: m	LR=m*l0, LR=1000
Electrode diameter dl: m	dl=0,04
Fault resolution time ts:(s)	ts=1
Weighting factor of the correction factor which adjusts the effects of Kii:	n0=1
Reference depth h0: (m)	h0=1

4. CALCULATION OF TOLERABLE TOUCH POTENTIAL

4.1.Surface layer derating factor Cs:

$$C_s := 1 - \frac{\left(0.09 \cdot \left(1 - \frac{p}{ps} \right) \right)}{2 \cdot h_s + 0.09}$$

$$C_s = 0.7827$$

4.2 Tolerable contact potential for a body weight of 50 & 70 kg:

Tolerable contact potential for a body weight of 50 kg :

$$TE_{touch50} := (1000 + 1.5 * C_s * ps) * 0.116 / \sqrt{ts}$$

$$TE_{touch50} = 1818.268 \text{ V}$$

Tolerable contact potential for a body weight of 70 kg :

$$TE_{touch70} = (1000 + 1.5 * C_s * ps) * 0.157 / \sqrt{ts}$$

$$TE_{touch70} = 2460.9327 \text{ V}$$

5.CALCULATION OF THE TOLERABLE STEP POTENTIAL

Tolerable step potential for 50 kg:

$$TE_{step50} = (1000 + 6.0 \cdot Cs \cdot ps) \cdot 0.116 / \sqrt{ts}$$

$$TE_{step50} = 6925.0751 \text{ V}$$

Tolerable step potential for 70 kg:

$$TE_{step70} = (1000 + 6.0 \cdot Cs \cdot ps) \cdot 0.157 / \sqrt{ts}$$

$$TE_{step70} = 9372.7309 \text{ V}$$

6. CALCULATION OF GROUNDING RESISTANCE

6.1 Coefficients of the Schwarz formula $ki1$:

$$Ki1 := 1.13 - 0.05 \cdot \frac{L}{B}$$

$$Ki1 = 1.0653$$

$$Ki2 := 4.4 - 0.05 \cdot \frac{L}{B}$$

$$Ki2 = 4.3353$$

6.2 Resistance of earth electrodes $Ré$:

$$R_{é} := \frac{\rho}{2 \cdot m \cdot \pi \cdot LR} \cdot \left(\ln \left(8 \cdot \frac{L}{R_{dl}} \right) - 1 + 2 \cdot Ki1 \cdot \ln \left(\frac{(\sqrt{m} - 1)^2}{\sqrt{Ag}} \right) \right)$$

$R_{é} =$

0.0474 Ohm,

6.3 Resistance of the grid to the ground:

$$R_s := \frac{\rho}{\pi \cdot L_c} \cdot \left(\ln \left(2 \cdot \frac{L_c}{\sqrt{d \cdot H}} \right) + \frac{Ki1 \cdot L_c}{\sqrt{Ag}} - Ki2 \right)$$

$$R_s = 4.8268 \text{ Ohm}$$

6.4 Mutual resistance between earth grid and earth electrodes:

$$R_m = \frac{\rho}{\pi \cdot L_c} \cdot \left(\ln \left(2 \cdot \frac{L_c}{10} \right) + \frac{K_{i1} L_c}{\sqrt{A_g}} - K_{i2} + 1 \right)$$

$R_m = 4.5059 \text{ Ohm}$,

6.5 Combined resistance of the earth grid and earth electrodes R_g :

$$R_g := \frac{R_e \cdot R_s - R_m^2}{R_e + R_s - 2 \cdot R_m}$$

$R_g = 4.8517 \text{ Ohm}$

According to equation 50 of the IEEE-80:2000 standard, the simplified calculation of the total resistance of the grounding system is:

$$R_g' := \frac{\rho}{4} \cdot \sqrt{\frac{\pi}{A_g}}$$

$R_g = 5.5525 \text{ Ohm}$,

6.6 Calculation of IG ground grid current :

$$G = D_f \times I_f \times S_f$$

I_G = Maximum network current

D_f = Decrement factor for the entire duration of the fault,

I_f = RMS value of the symmetrical earth fault current,

$D_f = 1,015$

$I_g = I_f/2 = 7500 \text{ A}$

S_f = Fault current division factor $S_f = 0.6$

$I_G = D_f I_f S_f = 9135 \text{ A}$

$GPR = I_G R_g = 44319.942 \text{ V}$

Where,

I_G = Maximum network current in A

R_g = Network resistance

GPR = Earth Potential Rise

After calculation we remarque $GPR > TE_{touch}$ and TE_{step} , furthermore, it is necessary to check the value of GPR, It must be higher than mesh voltage and the step voltage.

7. Calculation of mesh voltage E and step tension ES

7.1 According to equation 80 of IEEE 80 mesh voltage (IEEE80 - Eq.80)

$$E = K_m K_i \cdot IG / LM$$

Where LM = effective buried length applicable to grids earth electrodes and electrodes in the corners. As well as electrodes along the perimeter and inside the earth grid,

$$L_M = L_C + \left[1.55 + 1.22 \left(\frac{L_r}{\sqrt{L_x^2 + L_y^2}} \right) \right] L_R$$

Where:

L_C = total length of the network conductor in m

L_R = Total length of all earth electrodes, in meters

L_r (10)= length of each earth electrode, in meter

$L_x = L = 66$ m

$L_y = B = 51$ m

$$LM := L_C + \left(1.55 + 1.22 \cdot \left(\frac{10}{\sqrt{L_x^2 + L_y^2}} \right) \right) \cdot L_R$$

$LM = 4342.2678$ m

$$K_m = \frac{1}{2 \cdot \pi} \cdot \left[\ln \frac{D^2}{16 \cdot h \cdot d} + \frac{(D + 2 \cdot h)^2}{8 \cdot D \cdot d} - \frac{h}{4 \cdot d} \right] + \frac{K_{ii}}{K_h} \cdot \ln \left[\frac{8}{\pi(2 \cdot n - 1)} \right]$$

$$K_h := \sqrt{1 + \frac{H}{hO}}$$

$K_h = 1.3416$

h = depth of ground grid conductor = 1m,

ho = grid reference depth of 1m

Kii 1, for the grid with electrodes along of the perimeter of the grid,

Ki = irregularity factor = $0.644 + 0.148n$; $n = n_a \cdot n_b \cdot n_c \cdot n_d \dots$ Eqn. (7)

$n_a = 2 \cdot L_c / L_p$

$L_p = 2 \cdot (L_x + L_y)$, $L_p = 234$

$n_a = 2 \cdot L_c / L_p$, $n_a = 22.6154$

$$n_b := \sqrt{\frac{L_p}{4 \cdot \sqrt{A_g}}} \quad , \quad n_b = 1.0042$$

$n_c = 1$ for square and rectangular grids

$n_d = 1$ for square, rectangular and L grids

L_p = peripheral length of the grid in meters

L_c = total length of the conductors forming the grid in meters

D_s = grid spacing in meters

d = diameter of the grid conductor in meters

H = depth of burial in meters

$n = n_a \cdot n_b \cdot n_c \cdot n_d$

$n = 22.7093$

$K_i = 0.644 + 0.148 \cdot n$

$K_i = 4.005$

$$K_m = \frac{1}{2 \cdot \pi} \cdot \left(\ln \left(\frac{D_s^2}{16 \cdot H \cdot d} + \frac{(D_s + 2 \cdot H)^2}{8 \cdot D_s \cdot d} - \frac{H}{4 \cdot d} \right) + \frac{K_{ii}}{K_h} \cdot \ln \left(\frac{8}{\pi \cdot (2 \cdot n - 1)} \right) \right)$$

Either $E = \rho \cdot K_m \cdot K_i \cdot I_G / L$

$E = 2437.4065 \text{ V}$

7.2 According to equation 92 of IEEE-80 , the Step voltage will be:

$$ES = p \cdot K_s \cdot K_{i3} \cdot IG / L_s$$

$$\text{Where } L_s = 0.75 \cdot L_c + 0.85 \cdot L_R$$

$$K_{i3} = \text{irregularity factor} = 0.644 + 0.148 \cdot n$$

$$L_s = 0.75 \cdot L_c + 0.85 \cdot L_R$$

$$L_s = 2834.5$$

$$K_{i3} = 0.644 + 0.148 \cdot n$$

$$K_{i3} = 4.005$$

$$K_s := \frac{1}{p} \cdot \left(\frac{1}{2 \cdot H} + \frac{1}{D + H} + \frac{1}{D} \cdot (1 - 0.5^{n-2}) \right)$$

$$K_s = 0.2053$$

$$ES := p \cdot K_s \cdot K_{i3} \cdot IG / L_s$$

$$ES = 1926.2939 \text{ V}$$

Conclusion: The calculation of Touch voltage and Step voltage is correct and respect the condition of $GPR > TE_{\text{touch}}$ and $Testep$, Calculation meet the requirements of IEEE80:2000.

According to the calculation carried out we obtain the following results:

Resistivity p (Ohm-m)	727
Surface Resistivity ps (Ohm-m)	12500
Ground grid section A (mm ²)	95
Conductor spacing Ds (m)	2,5
Number of stakes (electrodes)	100

Lenth of stakes L0 (m)	10
Electrode diameter (m)	0,04
TEtouch70 (V)	2460,93
TEstep70 (V)	9372,73
Rg(Ohm)	4,85
E(V)	2437,41
ES(V)	1926,29

After calculating the geometric average of resistivity $p=727$ Ohm-m, we find that the number of electrodes is equal to 100, each 10m long. This solution is impractical and costly. To remedy this, we must reduce the earth resistivity as much as possible.

Solutions:

- 1. Earthing Improvement Embankment:**

Replace soil around electrodes with low resistivity (SEV1, layer3) material, particularly at SEV points 2,4, 5, 6 and 7.

- 2. Grounding Enhancement Fill:**

Replace soil around electrodes with low resistivity filler to achieve low ground resistance.

6.2.2 Variant 2: Using Lowest Resistivity from First Survey

Using soil of SEV1, layer 3, with a resistivity of $p= 65.02$ Ohm-m and a depth of 10.2m, to replace each higher soil resistivity such SEV2/layer1 , SEV4/layer1&2, SEV5/layer1&2, SEV6/layer3 and SEV7/layer1,2&4 .



		RESIS- TIVITE	deep- BY	Weighted average resistivity (*)	LOG(weighted average resistivity)
SON- DAGE TRANCHE (Ohm-m)			(m)	Ohm-m	Ohm-m
SEV1	1	54.3	1.5		
	2	283	3		
				158.28	5.06
	3	65.02	10.2		
	4	INFINITY			
SEV2	1	65.02	0.5		
	2	279.2	2		
				265.38	5.58
	3	268.6	22.5		
	4	INFINITY			
SEV3	1	982.3	0.5		
	2	807.3	10		
				815.63	6.70
	3	114.5			
SEV4	1	65.02	0.5		
	2	65.02	4		
				337.16	5.82
	3	512.1	7		
	4	INFINITY			
SEV5	1	65.02	0.65		
	2	65.02	15		
	3	27.2		65.02	4.17
SEV6	1	354.3	3.3		
	2	36.62	10.93		
	3	65.02		110.29	4.70
SEV7	1	65.02	2.65		
	2	65.02	7.3		
				200.74	5.30
	3	309.9	12.37		
	4	65.02			
			Arithmetic average	278.93	37.35
				LOG (Weighted Average Resistivity)	5.34
				Geometric Mean of Resistivity	207.62

The geometric average of resistivity is: 207,62 Ohm-m.

In this variant we will adjust the following parameters:

- number of electrode,
- length of electrode,
- diameter of electrode ,
- total number of electrodes,
- soil resistivity,
- conductor spacing

Soil resistivity P: (Ohm-m)	P=207,62
Resistivity of the upper layer ps: Ohm-m (typical soil to gravel value)	ps=5000
Electrode length l0 (m)	l0=3
Conductor spacing Ds: (m)	Ds=2,5
Depth of burial H (m)	H=0,8
No. of electrodes, Total number of electrodes m:	m=35
Electrode diameter dl: m	DI=0,02

After calculation , we get the following results:

Tolerable tactile potential $TE_{touch50} = 793.567 \text{ V}$

Tolerable tactile potential $TE_{touch70} = 1074.052 \text{ V}$

Tolerable step potential $TE_{step50} = 2826.27 \text{ V}$

Tolerable step potential $TE_{step70} = 3825.210 \text{ V}$

Combined resistance of the earth grid and earth electrodes

$R_g = 1.3792 \text{ Ohm}$,

Determination of Earth Potential Rise (GPR) : $GPR=12599.1255$
 $GPR > TE_{touch}$ and TE_{step} , it is necessary to check the calculation of the touch voltage and the step voltage.
 After calculation , we get the following results:

Resistivity p (Ohm-m)	207,26
Surface Resistivity p_s (Ohm-m)	5000
Ground grid section A (mm ²)	95
Conductor spacing D_s (m)	2,5
Number of stakes (electrodes)	35
Lenth of stakes L_0 (m)	3
Electrode diameter (m)	0,02
$TE_{touch70}$ (V)	1074.052
TE_{step70} (V)	3825.210
R_g (Ohm)	1,379
E (V)	1066.747
ES (V)	792.956

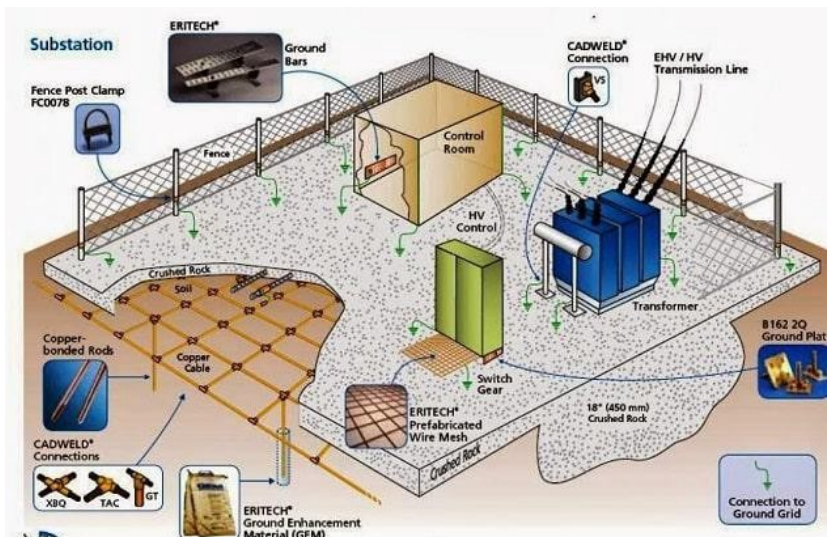
Conclusion: The calculation of Touch voltage and Step voltage is correct and respect the condition of $GPR > TE_{touch}$ and TE_{step} , Calculation meet the requirements of IEEE80:2000,

With number of electrode $m=35$, the length =3m and the diameter of electrode =20 mm , this solution is realizable and not expensive,

6.2.3 Variant 3: Additive Method of Improving Soil Conductivity

Using GEM earth enriched, a high conductivity material, we spread it around earth electrodes to lower earth resistance.

- **Affected Layers:** SEV2/layer1, SEV4&5/layer1&2, SEV6/layer3, SEV7/layer1,2&4
- **Resistivity:** 20 Ωm
- **Depth:** 3.5m, Diameter: 10 cm



		RESIST- QUICKLY	deep- BY		Weighted average resistivity (*)	LOG(weighted average resistivity)
SON- DAGE TRANCHE (Ohm-m)			(m)	Surface m²	Ohm-m	Ohm-m
SEV1	1	543	1.5	111.7		
	2	283	3			
					158.28	5.06
	3	65.02	10.2			
	4	INFINITY				
SEV2	1	20	0.5	111.7		
	2	279.2	2			
					264.48	5.58
	3	268.6	22.5			
	4	INFINITY				
SEV3	1	982.3	0.5	207.75		
	2	807.3	10			
					815.63	6.70
	3	114.5				
SEV4	1	20	0.5	102		
	2	20	4			
	3	512.1	7		319.54	5.77
	4	INFINITY				
SEV5	1	20	0.65	221.25		
	2	20	15			
	3	27.2			20.00	3.00
SEV6	1	354.3	3.3	173.7		
	2	36.62	10.93			
	3	20			110.29	4.70
SEV7	1	20	2.65	114		
	2	20	7.3			
					180.67	5.20
	3	309.9	12.37			
	4	20				
Arithmetic average					266.98	36.01
					LOG (Weighted Average Resistivity)	5.14
					Geometric Mean of Resistivity	171.41

In this variant we will adjust the following parameters:

- number of electrode,
- length of electrode,
- diameter of electrode ,
- total number of electrodes,
- soil resistivity,
- conductor spacing

Soil resistivity P: (Ohm-m)	P=171.41
Resistivity of the upper layer ps: Ohm-m (typical soil to gravel value)	ps=4000
Electrode length l0 (m)	l0=5
Conductor spacing Ds: (m)	Ds=2,5
Depth of burial H (m)	H=0,8
No. of electrodes, Total number of electrodes m:	m=25
Electrode diameter dl: m	Dl=0,02

After calculation , we get the following results:

- Tolerable tactile potential $TE_{touch50} = 658.2674 \text{ V}$
- Tolerable tactile potential $TE_{touch70} = 890.9309 \text{ V}$
- Tolerable step potential $TE_{step50} = 2285.0695 \text{ V}$
- Tolerable step potential $TE_{step70} = 3092.723 \text{ V}$
- Combined resistance of the earth grid and earth electrodes
 $R_g = 1.1395 \text{ Ohm}$,

Determination of Earth Potential Rise (GPR) : $GPR=10409.4364 \text{ V}$

$GPR > TE_{touch}$ and TE_{step} , it is necessary to check the calculation of the touch voltage and the step voltage.

After calculation , we get the following results:

$E_{(touch)}=882.709 \text{ V} < TE_{touch70}$

$ES_{(step)}=677.4046 \text{ V} < TE_{step70}$

Conclusion: The calculation of Touch voltage and Step voltage is correct and respect the condition of $GPR > TE_{touch}$ and TE_{step} , Calculation meet the requirements of IEEE80:2000,

With number of electrode $m=25$, the length $=5\text{m}$ and the diameter of electrode $=20 \text{ mm}$, this solution is realizable and shipper,

6.2.4 Conclusion

The design of the grounding network complies with IEEE 80-2000. Variants 2 and 3 are recommended due to their feasibility and cost-effectiveness.

7. Resources and Further Reading

Here's a detailed list of resources and further reading for deeper understanding in the field of substation earthing and electrical engineering:

Recommended Books

1. "Grounding and Shielding: Circuits and Interference" by Ralph Morrison"

- A comprehensive guide on grounding techniques and their importance in electrical systems.

2. IEEE Std 80-2000: IEEE Guide for Safety in Grounding of Substations"

- Essential standard for grounding practices in substations.

3. "Electrical Safety Handbook" by John Cadick, Ali Emadi, and David B. McRae

- Covers electrical safety principles and practices, including grounding.

4. "Power System Grounding" by David W. Jones

- Focuses on grounding principles specific to power systems.

5. "Substation Automation Systems: A Communication-Based Approach" by J. C. Das

- Details on automation and control in substations, including grounding considerations.

Articles and Journals

1. "Analysis of Grounding Systems in Electrical Substations" - IEEE Transactions on Power Delivery

- A peer-reviewed article discussing various grounding system analyses.

2. "Grounding of Power Systems" - CIGRÉ Technical Brochure

- Technical insights into the grounding of power systems and best practices.

3. "Review of Grounding Techniques for Substation Earthing" - Journal of Electrical Engineering

- A review article summarizing various grounding techniques.

Online Courses

1. Coursera - "Electrical Power Systems"

- An overview of power systems including grounding concepts.
- [Coursera Link](<https://www.coursera.org>)

2. edX - "Introduction to Electrical Engineering"

- Basic principles of electrical engineering, including grounding.
- [edX Link](<https://www.edx.org>)

3. Udemy - "Power System Analysis: A Practical Approach"

- Focuses on real-world applications of power system analysis.
- [Udemy Link](<https://www.udemy.com>)

Webinars and Workshops

1. IEEE Webinars on Power Systems

- Various topics including grounding and safety practices.
- [IEEE Link](<https://www.ieee.org>)

2. CIGRÉ Webinars

- Offers insights into international standards and practices in electrical engineering.
- [CIGRÉ Link](<https://www.cigre.org>)

3. NESC (National Electrical Safety Code) Workshops

- Focuses on safety standards and grounding practices.
- [NESC Link](<https://www.nesc.org>)

Additional Resources

- Technical Standards and Guidelines from IEEE, IEC, and ANSI
- Professional Organizations: Joining organizations such as IEEE Power and Energy Society can provide access to additional resources and networking opportunities.

8. Author's Note and Acknowledgments

Author's Note :

As the author of this book, I want to share my journey in the field of electrical engineering, specifically focusing on substation earthing. My fascination with electrical systems began at a young age, driven by a curiosity about how energy flows and powers our daily lives. Over the years, I have had the privilege of working on various projects that deepened my understanding of grounding systems and their critical role in ensuring safety and reliability in electrical infrastructure.

Writing this book has been a labor of love, motivated by a desire to share knowledge and best practices with fellow engineers, students, and enthusiasts. I believe that effective grounding is not just a technical requirement but a vital aspect of protecting lives and equipment. Through this work, I aim to bridge the gap between theory and practice, providing readers with practical insights and tools to enhance their understanding of substation earthing.

Acknowledgments :

I would like to extend my heartfelt thanks to those who contributed to this book and my journey:

- Industry Peers: A special thanks to my colleagues who shared their experiences and insights during the research phase of this book.

Your collaboration enriched the content and provided real-world context to the theories discussed.

- Professional Organizations: I am grateful to organizations such as IEEE, ASTM, IEC, CIGRÉ for their resources and standards that informed my work. Their commitment to advancing knowledge in electrical engineering is commendable.

- Family and Friends: Lastly, to my family and friends, thank you for your unwavering support and understanding during the long hours spent writing. Your belief in my vision kept me motivated.

This book is a culmination of collective knowledge and experience, and I hope it serves as a valuable resource for all who read it. Together, we can continue to advance the field of electrical engineering and ensure a safe and efficient future for our energy systems.

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